SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

Code of Conduct:

I Pranesh.R C191226039 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Problem Identification	20%	1	1	1	1	1
Debugging	25%	1	1	1	1	1
Optimization	20%	1	1	1	1	1
Analysis	20%	1	1	1	1	1
Code Quality	15%	1	1	1	1	2

24/30

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	**Next Hop**
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

SIMATS ENGINEERING

ASSESSMENT TOOL 3 – ANSWERS

COURSE: COMPUTER NETWORKS | COURSE CODE: CSA07

Name: Pranesh.R

Register Number: 1912260397

Course Outcome: CO2 – Configure IP addressing schemes, subnetting, and basic routing techniques.

Q1. Debugging IPv4 Addressing Configuration

1. Given configuration

- IP address: 192.168.10.65
- Subnet mask: 255.255.255.192 (/26)
- Configured default gateway: 192.168.10.1

2. Identify the subnet

A /26 subnet has 64 addresses per block because 256 − 192 = 64. The subnet blocks in the last octet are 0–63, 64–127, 128–191, and 192–255. Therefore, 192.168.10.65 belongs to the subnet 192.168.10.64/26, not 192.168.10.0/26.

3. Network details

- Network address: 192.168.10.64
- Broadcast address: 192.168.10.127
- Valid host range: 192.168.10.65 to 192.168.10.126
- Total addresses: 64; usable host addresses: 64 − 2 = 62

4. Identify the errors

The IP address 192.168.10.65 is valid for subnet 192.168.10.64/26. However, the configured gateway 192.168.10.1 belongs to 192.168.10.0/26, so it is outside the workstation's subnet. Also, the workstation is not in the stated target subnet 192.168.10.0/26.

5. Corrected configuration

- If the workstation must communicate in 192.168.10.0/26: IP address 192.168.10.10
- Subnet mask: 255.255.255.192
- Default gateway: 192.168.10.1 (provided that this address is configured on the router interface for this subnet)

The corrected subnet is 192.168.10.0/26. Its network address is 192.168.10.0, broadcast address is 192.168.10.63, valid host range is 192.168.10.1–192.168.10.62, and usable host count is 62.

Conclusion: The main problem is a subnet mismatch between the workstation IP and the intended network. Assigning an address from 192.168.10.0/26 and using a gateway in the same subnet corrects the configuration.

Q2. Debugging IPv6 Addressing

1. Given addresses

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

2. Expand the addresses

Device A: 2001:0db8:0000:0000:0000:ff00:0042:8329/64

Device B: 2001:0db8:0000:0000:0001:0000:0000:0001/64

3. Network prefix and subnet check

With a /64 prefix, the first four hextets (64 bits) identify the network.

- Device A prefix: 2001:0db8:0000:0000::/64
- Device B prefix: 2001:0db8:0000:0000::/64

Both addresses share the same first four hextets. Therefore, despite the apparent extra '1' in Device B's address after the compressed zeros, the correct expansion is 2001:0db8:0000:0000:0001:0000:0000:0001, whose /64 prefix is 2001:db8:0:0::/64. The devices are in the same /64 subnet.

4. Debugging and optimization

There is no /64 network-prefix mismatch in the addresses as written. If communication fails, check that both devices have valid interface configuration, are on the same Layer 2 network or have correct IPv6 routing, and that ICMPv6/neighbor discovery is not blocked. For a simple same-subnet example, Device A may use 2001:db8:0:0::10/64 and Device B may use 2001:db8:0:0::20/64.

5. Number of addresses

An IPv6 /64 subnet contains 2^(128−64) = 2^64 addresses. This is 18,446,744,073,709,551,616 addresses. IPv6 does not use the IPv4-style network and broadcast address subtraction; the subnet has 2^64 possible interface identifiers.

Conclusion: Both provided addresses have the same /64 prefix. The fault cannot be confirmed as an addressing-prefix mismatch from the supplied information; further link, routing, and device checks are needed.

Q3. Optimize IPv4 Subnet Allocation Using VLSM

1. Given information

The organization has 192.168.1.0/24 and uses /26, /27, and /28 subnets. The question does not provide the number of hosts required by each department or the exact current subnet assignments. Therefore, the exact unused-address count and specific existing allocation mistakes cannot be calculated without making assumptions.

2. Capacity of each subnet size

Prefix	Subnet mask	Total addresses	Usable hosts	Typical use
/26	255.255.255.192	64	62	Up to 62 hosts
/27	255.255.255.224	32	30	Up to 30 hosts
/28	255.255.255.240	16	14	Up to 14 hosts

3. Example subnet boundaries

Subnet	Network	Valid host range	Broadcast	Usable
/26	192.168.1.0	192.168.1.1–192.168.1.62	192.168.1.63	62
/26	192.168.1.64	192.168.1.65–192.168.1.126	192.168.1.127	62
/27	192.168.1.128	192.168.1.129–192.168.1.158	192.168.1.159	30
/27	192.168.1.160	192.168.1.161–192.168.1.190	192.168.1.191	30
/28	192.168.1.192	192.168.1.193–192.168.1.206	192.168.1.207	14
/28	192.168.1.208	192.168.1.209–192.168.1.222	192.168.1.223	14

The remaining range 192.168.1.224–192.168.1.255 can be allocated as one /27 subnet (network 192.168.1.224, usable hosts 192.168.1.225–192.168.1.254, broadcast 192.168.1.255), if required.

4. VLSM optimization method

1. List each department's actual host requirement, including expected growth.
2. Sort departments from the largest host requirement to the smallest.
3. Choose the smallest subnet that supports each requirement: /26 supports 62 hosts, /27 supports 30, and /28 supports 14.
4. Allocate non-overlapping subnets on valid block boundaries and record each network, host range, and broadcast address.
5. Reserve spare subnets for growth and verify that no two departments use overlapping address ranges.

5. How to identify unused addresses

For each department, unused usable addresses = usable host capacity − required host addresses. For example, if a department needs 20 hosts and receives a /27, it has 30 usable addresses and 10 remain unused. If it receives a /26 instead, 42 usable addresses remain unused. Actual totals require the host counts and current assignments.

Conclusion: VLSM improves address use by matching subnet size to each department's requirement. The supplied task lacks departmental host counts, so the table above is an illustrative valid allocation, not a claim about the organization's existing layout.

Q4. IPv6 Address Debugging (Repeated Task)

This question repeats Question 2, so the same analysis applies.

- Device A expanded: 2001:0db8:0000:0000:0000:ff00:0042:8329/64
- Device B expanded: 2001:0db8:0000:0000:0001:0000:0000:0001/64
- Both /64 network prefixes: 2001:0db8:0000:0000::/64
- Both devices are in the same /64 subnet.
- Example optimized addresses: Device A 2001:db8:0:0::10/64; Device B 2001:db8:0:0::20/64.
- Available addresses in a /64: 2^64 = 18,446,744,073,709,551,616.

If communication still fails, verify link status, IPv6 configuration, neighbor discovery, firewall rules, and routing. The given prefixes alone do not prove an address mismatch.

Q5. Debugging and Optimizing the Routing Table

1. Given routing table

Destination network	Next hop	Prefix length
192.168.1.0/24	10.0.0.2	/24
192.168.2.0/24	10.0.0.3	/24
Default route (0.0.0.0/0)	10.0.0.1	/0

2. Longest-prefix match for 192.168.2.100

The destination 192.168.2.100 belongs to 192.168.2.0/24 because the first 24 bits match that network. The router selects the 192.168.2.0/24 entry and forwards the packet to next hop 10.0.0.3. This /24 route is more specific than the default /0 route.

3. Verify destination subnet

For 192.168.2.0/24, the network address is 192.168.2.0, broadcast address is 192.168.2.255, and valid host range is 192.168.2.1–192.168.2.254. Therefore, 192.168.2.100 is a valid host in the intended subnet.

4. Possible routing errors and debugging

- Check that next hop 10.0.0.3 is reachable from the router and is on a directly connected or otherwise reachable network.
- Verify that the next-hop router has a route to 192.168.2.0/24 and that the destination LAN has a return route.
- Check interface status, subnet masks, access-control lists, and firewall rules.
- Use Packet Tracer simulation mode or ping/traceroute to identify where packets stop.

5. Optimization and default route

The listed table already contains a specific /24 route for 192.168.2.0/24 and a default route. Keep the specific route if 10.0.0.3 is the correct reachable next hop. Remove or correct duplicate, stale, or incorrect routes if found; do not change the next hop without confirming the topology.

If no more-specific route matches a destination, the router uses the default route 0.0.0.0/0 via 10.0.0.1. If no default route is installed, the router drops the packet and may return an ICMP destination-unreachable message.

Conclusion: Traffic for 192.168.2.100 should use next hop 10.0.0.3. Since the routing-table entry is logically correct, the failure may be caused by next-hop reachability, downstream routing, a return-path issue, or filtering.

Overall Conclusion

This assessment demonstrates IPv4 subnet calculations, IPv6 prefix checking, VLSM planning, and longest-prefix routing. A correct configuration must use valid host addresses, matching subnet prefixes, non-overlapping subnet allocations, reachable next hops, and a working return path. Where the task does not supply host requirements or network topology, those details must be verified rather than assumed.